

ResQGrid AI — Business Case Document

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GitHub: <https://github.com/deshanlakshitha/resqgrid-ai-v1>

Live app: <https://web-two-mu-edisuy5tdl.vercel.app>

API: <https://resqgrid-api-9wns.onrender.com/api/v1>

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1. Problem Statement

Sri Lanka experiences severe flooding almost every monsoon season. In November 2025 alone, floods and mudslides killed more than 330 people (BBC). In 2016, over 300,000 people were displaced (IOM/UN sitreps). The World Bank documents forty consecutive years of annual flood damage across the island.

At the centre of every district response sits a single duty officer at the District Emergency Operations Centre (DMC). During a flood, reports reach this officer as WhatsApp voice notes, phone calls, and radio chatter — unstructured, untriaged, and competing for attention. The officer must answer six critical questions simultaneously:

- Which incidents are most urgent?
- Who is at risk?
- What resources are available right now?
- Which resource should go where?
- Which routes are still passable?
- What changed since my last decision?

Existing tools — static asset registries, group chats, paper logs — record information but do not understand, rank, or act on it. The gap is not data collection; it is real-time decision support.

2. Proposed Solution

ResQGrid AI is an intelligent emergency resource coordination system. Its operating principle is: AI recommends. Humans approve. Every important decision is explainable and auditable.

The platform ingests incident reports in plain language from citizens, field responders, and operators. An AI triage pipeline (powered by Alibaba Cloud Model Studio / Qwen) structures each report into a typed incident schema, extracts severity indicators, and supplies the data to a deterministic priority engine. The engine produces a numeric score that decomposes into six explainable weighted factors. A resource-matching module then proposes the optimal available unit considering type, proximity, hazard zones, and route risk. A human dispatcher reviews the recommendation and approves every dispatch — the system cannot move a vehicle without authorization.

Key design decisions:

- AI never writes directly to the database — all output is validated against strict Pydantic schemas.
- Priority scoring is fully deterministic and auditable — every point is traceable to a factor.
- Human-in-the-loop is enforced in code, not just policy — approval is a required API step.

- Graceful degradation — if the AI provider is unavailable, the system falls back to rule-based triage.
- Offline-first incident reporting — reports are queued in IndexedDB and replayed on reconnect.

3. Target Audience

User type	Role	Pain point addressed
DMC Duty Officer	Primary user — approves all dispatches	Replaces paper logs and phone triage with a structured, prioritized interface
Field Responder	Accepts and advances assignments	Receives clear, approved task details on mobile; no ambiguity
Citizen / Affected person	Reports incidents	Plain-language reporting via mobile web; no app install required
District Coordinator	Oversight and audit	Full audit log of every AI output and human decision

4. Value Proposition

ResQGrid AI compresses the triage cycle from minutes to seconds. A duty officer previously had to listen to a voice note, mentally categorize the incident, locate it on a paper map, call resource units to check availability, and make a verbal dispatch decision. Each step is manual, undocumented, and unrepeatable.

With ResQGrid AI:

- Incident reports are structured and classified in under 2 seconds by AI.
- Priority scores are calculated instantly with an explainable breakdown.
- The best available resource is surfaced with a confidence score.
- The dispatcher approves with one click; the assignment is logged immutably.
- If a route is blocked, the system re-evaluates and proposes an alternative.

Compared to competitors: generic incident-management platforms (e.g., Sahana Eden, WebEOC) are data-entry systems, not decision engines. They require manual categorization and provide no AI-assisted prioritization or resource optimization. ResQGrid AI is the first open-source, AI-native, human-supervised emergency dispatch prototype designed specifically for the Sri Lankan district DMC workflow.

5. Business Model and Feasibility

5.1 Current deployment cost

The live demo runs on free-tier cloud infrastructure: Vercel (frontend), Render (backend), Supabase (PostgreSQL), Redis Cloud (cache), and Alibaba Cloud Model Studio (pay-per-token AI). Total ongoing cost for the prototype: approximately USD 0 per month.

5.2 Path to real implementation

Phase	Milestone	Timeline
Pilot	Deploy at one district DMC office with fictional data; train 2 duty officers	3 months
Integration	Connect to the National Disaster Relief Services Centre (NDRSC) data feeds; add Sinhala/Tamil intake	6 months
Scale	Deploy across all 25 districts; integrate IoT flood sensors for re-planning triggers	12-18 months
Sustain	Government licensing or NGO grant funding; long-term maintenance contract	18 months+

5.3 Revenue / funding model

- Government SaaS licence (per-district annual fee)
- UNDP / World Bank humanitarian innovation grants
- NGO partnership deployment (ICRC, Oxfam, CARE Sri Lanka)
- Open-source community edition; paid enterprise support

6. Competition and Differentiation

Platform	AI triage	Human approval loop	Open source	Sri Lanka fit
Sahana Eden	No	No	Yes	Partial
WebEOC	No	No	No	No
Ushahidi	No	No	Yes	Partial
ResQGrid AI	Yes	Yes	Yes	Designed for it

7. Technical Feasibility Summary

Component	Technology	Status
Frontend	Next.js 14 / TypeScript / Tailwind CSS / Mapbox	Deployed on Vercel
Backend API	Python 3.11 / FastAPI / SQLAlchemy / Alibaba Cloud	Live on Render
Database	PostgreSQL 16 + PostGIS 3.4	Live on Supabase
Cache / Queue	Redis 7	Live on Redis Cloud
AI / LLM	Alibaba Cloud Model Studio — Qwen-plus	Integrated
Object Storage	Alibaba Cloud OSS (S3-compatible)	Integrated
Mobile	PWA + Capacitor Android APK	Built
Auth	JWT + RBAC (4 roles)	Implemented
Deployment	Docker Compose / Vercel / Render	Automated

8. Risk Assessment

Risk	Likelihood	Mitigation
AI provider downtime	Low	Rule-based fallback triage engine always active
Internet outage in field	Medium	Offline IndexedDB outbox; reports sync on reconnect
LLM hallucination	Medium	Pydantic schema validation rejects malformed AI output
Adoption resistance	Medium	Human-approval design preserves officer authority; no automa
Data privacy	Low	Demonstration uses fictional data; production would use nation

9. Conclusion

ResQGrid AI demonstrates that a practical, affordable, AI-assisted emergency coordination system can be built and deployed in weeks using open-source components and cloud AI. The live

prototype is running today. The next step is a government pilot. With 330+ deaths in the last major Sri Lanka flood event, faster, smarter triage is not a nice-to-have — it is a life-safety imperative.

"AI recommends. Humans approve. Every important decision is explainable and auditable."